

Technical Document #372: Cybersecurity - Comprehensive Overview

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Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Security information and event management (SIEM) systems collect and analyze security logs.
- Multi-factor authentication (MFA) requires multiple forms of verification.
- Zero trust architecture assumes no implicit trust and verifies every request.